

CIRCLE DIFFEOMORPHISMS

[after M. R. HERMAN]

by Pierre DELIGNE

Let f be an orientation-preserving diffeomorphism of the circle $\mathbf{R}/\mathbf{Z}$.

If the rotation number $\alpha \in \mathbf{R}/\mathbf{Z}$ of f is irrational (i.e. if f has no periodic point) and the derivative Df of f has bounded variation (i.e. if the second derivative D^2f —taken in the sense of distributions—is a measure), Denjoy proved that f is topologically conjugate to the rotation

$$R_\alpha : x \mapsto x + \alpha :$$

there exists an orientation-preserving homeomorphism $h : \mathbf{R}/\mathbf{Z} \rightarrow \mathbf{R}/\mathbf{Z}$ such that $hfh^{-1} = R_\alpha$. This homeomorphism is unique up to composition on the left by a rotation, because the centralizer of an irrational rotation consists only of rotations.

It amounts almost to the same thing to prove a regularity property for h , or the same property uniformly for all iterates f^n of f . To pass from h to the f^n , one writes

$$f^n = (h^{-1}R_\alpha h)^n = h^{-1}R_{n\alpha}h.$$

Conversely, it is useful to change notation and pass to the universal cover $\mathbf{R}$ of the circle $\mathbf{R}/\mathbf{Z}$: f becomes an increasing diffeomorphism of $\mathbf{R}$, and h an increasing homeomorphism of $\mathbf{R}$, defined up to addition of a constant; both are of the form $x + \varphi(x)$, with φ periodic of period 1, and

$$hfh^{-1} = R_\alpha : x \mapsto x + \alpha,$$

with $\alpha \in \mathbf{R}$ (the rotation number). Set

$$h_n = \frac{1}{n} \sum_{i=0}^{n-1} (f^i - i\alpha).$$

We shall see that h , suitably normalized, is the limit of the h_n (3.1). Suppose now that the f^n are uniformly $C^{r+\beta}$, i.e. that f is C^r and that the r -th derivatives $D^r f^n$ satisfy, uniformly in n , a Hölder condition of exponent β ($0 < \beta \leq 1$):

$$|D^r f^n(x) - D^r f^n(y)| \leq A|x - y|^\beta.$$

The $f^i - R_{i\alpha}$ satisfy the same condition and are, moreover, bounded (by one). The same is true of the $h_n - \text{Id}$, which are their barycenters. Thus the $h_n - \text{Id}$ form a relatively compact set of functions on $\mathbf{R}/\mathbf{Z}$ in the C^r topology, and one can extract a convergent subsequence from the sequence h_n . The conjugacy h is therefore C^r , and passage to the limit shows that $D^r h$ satisfies the same Hölder condition as the $D^r f^n$.

Herman proved a theorem slightly stronger than the following.

THEOREM.— If the continued-fraction expansion of α has bounded coefficients and f is C^∞ , then h is C^∞ .

First (§§ 1 to 6), one proves that, as q ranges over the denominators of the convergents p_n/q_n of the continued-fraction expansion of α , $|Df^q|$ tends rapidly to 0 (at rate $1/q^\varepsilon$, with $\varepsilon > 0$). Such a result is plausible: the rotation $R_{q\alpha}$ is close to the identity, and $f^q = h^{-1}R_{q\alpha}h$.

It follows that h is $C^{1+\varepsilon'}$, with $\varepsilon' > 0$; then, from the fact that f is $C^{1+\varepsilon'}$ -conjugate to R_α , one deduces that the conjugates $f_n = h_n f h_n^{-1}$ of f tend to R_α in the $C^{2+\varepsilon''}$ topology ($\varepsilon'' > 0$) (§§ 7 and 8).

An implicit-function theorem of Arnold-Moser ([1], [3]), improved by Herman following Rüssmann [4], finally shows that if f is sufficiently close to R_α in the $C^{2+\varepsilon}$ topology ($\varepsilon > 0$), then f is C^∞ -conjugate to a rotation.

Terminology and notation

Write $\|\psi\|_\infty$ and $\|\psi\|_1$ for the L^∞ and L^1 norms of a function ψ on $\mathbf{R}/\mathbf{Z}$: $\sup |\psi(x)|$ and $\int_{\mathbf{R}/\mathbf{Z}} |\psi(x)| dx$. For $0 \leq \beta \leq 1$, write $|\psi|_\beta$ for its Hölder norm of exponent β :

$$|\psi|_\beta = \sup_{x \neq y} \frac{|\psi(x) - \psi(y)|}{|x - y|^\beta}.$$

Write $\text{Var}(\psi)$ for the total variation of ψ . If ψ is C^1 , this is $\|D\psi\|_1$.

R_s = the “rotation” $x \mapsto x + s$.

$f : \mathbf{R} \rightarrow \mathbf{R}$ is an increasing diffeomorphism of the form $x + \varphi(x)$, with φ periodic of period 1. Its iterates are denoted by f^n ($n \in \mathbf{Z}$). By hypothesis, f and f^{-1} are C^1 ; in addition, Df (and hence $\log Df$) is assumed to have bounded variation.

$$V = \text{Var}(\log Df).$$

α = the rotation number of f . We have $\|f^n - R_{n\alpha}\|_\infty < 1$. We assume that α is irrational.

μ = the probability measure on $\mathbf{R}/\mathbf{Z}$ invariant under f . If h conjugates f to R_α , it carries μ to dx : $\mu = dh$ (the distributional derivative, or Stieltjes integral).

h = the increasing homeomorphism of $\mathbf{R}/\mathbf{Z}$ conjugating f to R_α , normalized by the condition

$$\int (h(x) - x) \mu = 0.$$

$$h_n = \frac{1}{n} \sum_{i=0}^{n-1} (f^i - i\alpha).$$

This is a barycenter of increasing diffeomorphisms, hence an increasing diffeomorphism. We have $h = \lim h_n$ (3.1).

$$f_n = h_n f h_n^{-1}.$$

A rational approximation to $s \in \mathbf{R}$ is an irreducible fraction p/q such that $|s - p/q| < 1/q^2$, i.e. $|qs - p| < 1/q$. Every irrational number has infinitely many rational approximations (Dirichlet).

q denotes the denominator of a rational approximation to α (and p its numerator).

§ 1. First bound for $|Df^q|$; the Denjoy-Koksma inequality

THEOREM 1.1 (Denjoy-Koksma).— Let $s \in \mathbf{R}$, let m be the denominator of a rational approximation to s , and let ψ be a function of bounded variation on $\mathbf{R}/\mathbf{Z}$. Then

$$\left| \sum_{i=0}^{m-1} \psi(x + is) - m \int_{\mathbf{R}/\mathbf{Z}} \psi(t) dt \right| \leq \text{Var}(\psi).$$

After the change of variable $t \mapsto \pm(t - x)$, it suffices to assume that $x = 0$ and $n/m \leq s \leq n/m + 1/m^2$ (with $(n, m) = 1$). For $0 \leq i \leq m - 1$, we have $in/m \leq is < in/m + 1/m$, and the points in/m run through $\frac{1}{m}\mathbf{Z}/\mathbf{Z}$. Hence (the sums run from $i = 0$ to $i = m - 1$)

$$\sum_{i=0}^{m-1} \psi(is) - m \int_{\mathbf{R}/\mathbf{Z}} \psi(x) dx = \sum_{i=0}^{m-1} \left(\psi(is) - m \int_{in/m}^{in/m+1/m} \psi(x) dx \right),$$

and

$$\left| \psi(is) - \int_{in/m}^{in/m+1/m} \psi(x) (m dx) \right|$$

is bounded by the variation of ψ between in/m and $in/m + 1/m$, proving the theorem.

Recall that the variation $\int_{\mathbf{R}/\mathbf{Z}} |d\psi|$ is invariant under change of variables. Take $s = \alpha$, and make the change of variable that conjugates the rotation R_α to f ; it carries dx to the measure μ , and one obtains:

COROLLARY 1.2.— For a function ψ of bounded variation on $\mathbf{R}/\mathbf{Z}$,

$$\left| \sum_{i=0}^{q-1} \psi \circ f^i - q \int_{\mathbf{R}/\mathbf{Z}} \psi(x) \mu \right| \leq \text{Var}(\psi).$$

1.3. We have

$$Df^n = Df \circ f^{n-1} \cdot Df^{n-1} = Df \circ f^{n-1} \cdot Df \circ f^{n-2} \cdots Df,$$

whence

$$\log Df^n = \sum_{i=0}^{n-1} \log Df \circ f^i. \quad (1.3.1)$$

Applying the bound 1.2 with $\psi = \log Df$ gives

$$\left| \log Df^q - q \int_{\mathbf{R}/\mathbf{Z}} \log Df \cdot \mu \right| \leq V.$$

If the integral were nonzero, $\log Df^q$ would tend uniformly to $\pm\infty$ as $q \rightarrow \infty$, and Df^q would tend uniformly to 0 or ∞ . Since

$$\int_{\mathbf{R}/\mathbf{Z}} Df^q(x) dx = 1,$$

this is impossible, and therefore:

COROLLARY 1.4

$$\int_{\mathbf{R}/\mathbf{Z}} \log Df \cdot \mu = 0.$$

COROLLARY 1.5

$$e^{-V} \leq Df^q \leq e^V \quad (\text{when } |\alpha - p/q| < 1/q^2).$$

Remark 1.6.— The proof method of Theorem 1.1 proves 1.2 directly for every increasing homeomorphism f (and for μ an invariant measure). If $V = \int |d \log Df| < \infty$, one obtains 1.5. In his note [2], Denjoy starts from 1.5 to prove that f is topologically conjugate to a rotation.

1.7. Let s be irrational, let q_i ($i \geq 0$) be an increasing sequence of rational approximations to s , and let a_i be the integer part of q_i/q_{i-1} ($i \geq 1$). Assume that $q_0 = 1$. Every integer $N < q_n$ can be written

$$N = \sum_{i < n} b_i q_i \quad \text{with } b_i \leq a_{i+1} :$$

if $n = 0$, this is clear; if $n > 0$, write $N = b_{n-1} q_{n-1} + N'$ with $N' < q_{n-1}$, and conclude by induction. The sum $\sum_{i < N} \psi(x + is)$ then decomposes into $(\sum b_i)$ partial sums, each having one of the q_i as its length, and (1.1) gives

$$\left| \sum_{i=0}^{N-1} \psi(x + is) - N \int_{\mathbf{R}/\mathbf{Z}} \psi(t) dt \right| \leq \sum_i b_i \text{Var}(\psi) \leq \sum_{q_i \leq N} a_{i+1} \text{Var}(\psi), \quad (1.7.1)$$

and, for $s = \alpha$, (1.2) and (1.3) give

$$\left\| \sum_{i=0}^{N-1} \psi \circ f^i - N \int \psi \mu \right\|_{\infty} \leq \sum_{q_i \leq N} a_{i+1} \text{Var}(\psi), \quad (1.7.2)$$

$$\| \log Df^N \|_{\infty} \leq \sum_{q_i \leq N} a_{i+1} V. \quad (1.7.3)$$

§ 2. Bound for $|D^2 f^q|$

2.1. Differentiate formula (1.3.1) twice:

$$\log Df^n = \sum_{i < n} \log Df \circ f^i :$$

$$D \log Df^n = \sum_{i < n} D \log Df \circ f^i \cdot Df^i, \quad (2.1.1)$$

that is,

$$D^2 f^n = Df^n \cdot \sum_{i < n} D \log Df \circ f^i \cdot Df^i. \quad (2.1.2)$$

$$\begin{aligned}
 D^2 \log Df^n &= \sum_{i < n} (D^2 \log Df \circ f^i \cdot (Df^i)^2 + D \log Df \circ f^i \cdot D^2 f^i) \\
 &= \sum_{i < n} D^2 \log Df \circ f^i \cdot (Df^i)^2 + \sum_{j < i < n} D \log Df \circ f^i \cdot Df^i \cdot D \log Df \circ f^j \cdot Df^j.
 \end{aligned} \tag{2.1.3}$$

In the last term one recognizes (Oh! miracle) the cross terms in the expansion of the square of the right-hand side of (2.1.1). Completing the square gives

$$D^2 \log Df^n = \frac{1}{2} (D \log Df^n)^2 - \frac{1}{2} \sum_{i < n} (D \log Df \circ f^i \cdot Df^i)^2 + \sum_{i < n} D^2 \log Df \circ f^i \cdot (Df^i)^2. \tag{2.1.4}$$

If in (2.1.4) one bounds the $|Df^i|$ by $\sup_{i < n} \|Df^i\|_\infty$, one obtains

$$\|D^2 \log Df^n\|_\infty \leq \frac{1}{2} \|D \log Df^n\|_\infty^2 + \frac{1}{2} A n \sup_{i < n} \|Df^i\|_\infty^2, \tag{2.1.5}$$

where

$$A = \|D \log Df\|_\infty^2 + 2\|D^2 \log Df\|_\infty. \tag{2.1.6}$$

2.2. By Hadamard, if ψ is a C^2 function on $\mathbf{R}/\mathbf{Z}$, then (cf. 7.1.4)

$$\|D\psi\|_\infty \leq (2\|\psi\|_\infty \|D^2\psi\|_\infty)^{1/2}. \tag{2.2.1}$$

Assume that f is C^3 , and apply this interpolation inequality to $\psi = \log Df^q$, taking account of (1.5) and (2.1.5):

$$\|D \log Df^q\|_\infty \leq \left(V \|D \log Df^q\|_\infty^2 + q A V \sup_{i < q} \|Df^i\|_\infty^2 \right)^{1/2}. \tag{2.2.2}$$

If $V < 1$, this gives an estimate for $\|D \log Df^q\|_\infty$ (which occurs on both sides of the formula): squaring, one finds

$$\|D \log Df^q\|_\infty^2 \leq \frac{AV}{1-V} \cdot q \cdot \sup_{i < q} \|Df^i\|_\infty^2.$$

Writing $D \log Df^q = D^2 f^q / Df^q$, and applying 1.5 once again, one finally obtains:

THEOREM 2.3.— Assume that f is C^3 and that $V < 1$. Let

$$B = \left(\frac{AV}{1-V} \right)^{1/2} \cdot e^V,$$

where A is defined by (2.1.6). If q is the denominator of a rational approxi-

mation to α , then

$$\|D^2 f^q\|_\infty \leq B q^{1/2} \sup_{i < q} \|D f^i\|_\infty.$$

The essential gain in this estimate is that a factor $\sqrt{q}$, rather than q , has been obtained.

§ 3. $\text{Var}(\log D f_n)$ tends to 0

In order to apply 2.3, it is necessary to reduce to the case $V < 1$. Later, in order to control the $\|D f^i\|_\infty$, ($i < q$), we shall even need to reduce to the case in which V is small. The main result 3.3 of this section will make this possible.

PROPOSITION 3.1.— The h_n converge uniformly to h .

Put $\psi(x) = h(x) - x$. By definition, h is normalized so that

$$\int_{\mathbf{R}/\mathbf{Z}} \psi(x) \mu = 0.$$

The formula $R_\alpha h = h f$ may be written $h = h \circ f - \alpha = f - \alpha + \psi \circ f$. Likewise, since h conjugates f^i to $R_{i\alpha}$, we have

$$h = f^i - i\alpha + \psi \circ f^i.$$

Take the mean of these identities for $0 \leq i < n$:

$$h = \frac{1}{n} \sum_{i < n} (f^i - i\alpha) + \frac{1}{n} \sum_{i < n} \psi \circ f^i = h_n + \frac{1}{n} \sum_{i < n} \psi \circ f^i.$$

It remains to show that $\frac{1}{n} \sum_{i < n} \psi \circ f^i$ converges uniformly to $\int \psi \mu = 0$. This can be deduced from 1.7 (ψ has variation ≤ 2), or, by a change of variable (using h), this convergence can be reduced to a theorem of H. Weyl: Reminder 3.2 (H. Weyl).— Let φ be a continuous function on $\mathbf{R}/\mathbf{Z}$, and let α be irrational. The sums

$$S_n(\varphi) = \frac{1}{n} \sum_{i < n} \varphi(x + i\alpha)$$

converge uniformly to

$$\int_{\mathbf{R}/\mathbf{Z}} \varphi(x) dx.$$

This is seen by approximating φ by trigonometric polynomials P . One checks directly that, for a trigonometric polynomial, $S_n(P)$ tends to $\int_{\mathbf{R}/\mathbf{Z}} P(x) dx$. Since $|S_n(\varphi)| \leq \|\varphi\|_\infty$, it follows that

$$\begin{aligned} \limsup \left\| S_n(\varphi) - \int_{\mathbf{R}/\mathbf{Z}} \varphi(x) dx \right\|_{\infty} &= \limsup \left\| S_n(\varphi - P) - \int_{\mathbf{R}/\mathbf{Z}} (\varphi(x) - P(x)) dx \right\|_{\infty} \\ &\leq 2\|\varphi - P\|_{\infty}, \end{aligned}$$

which is arbitrarily small.

THEOREM 3.3.— If f is C^2 , the variation

$$V_n = \|D \log Df_n\|_1$$

tends to 0 as $n \rightarrow \infty$.

Let us compute Df_n . We have

$$\begin{aligned} h_n \circ f &= \frac{1}{n} \sum_{i=0}^{n-1} (f^{i+1} - i\alpha) = \frac{1}{n} \sum_{i=1}^n (f^i - i\alpha) + \alpha \\ &= h_n + \alpha + \frac{1}{n} (f^n - x - n\alpha), \end{aligned}$$

whence

$$f_n = R_{\alpha} + \frac{1}{n} (f^n - x - n\alpha) \circ h_n^{-1}$$

and

$$Df_n = 1 + \left[\frac{1}{n} (Df^n - 1) \cdot (Dh_n)^{-1} \right] \circ h_n^{-1}.$$

Thus

$$Df_n - 1 = \frac{Df^n - 1}{\sum_{i < n} Df^i} \circ h_n^{-1}.$$

Furthermore, since $Df^n = Df \circ f^{n-1} \cdot Df^{n-1}$ and

$$(Df^{n-1})^{-1} \circ f^{1-n} = D(f^{1-n}),$$

we have

$$\frac{Df^n}{\sum_{i < n} Df^i} \circ f^{1-n} = \frac{Df}{\sum_{i < n} Df^i \circ f^{1-n} \cdot D(f^{1-n})} = \frac{Df}{\sum_{i < n} Df^{-i}},$$

that is,

$$Df_n - 1 = \left(\frac{Df}{\sum_{i < n} Df^{-i}} \circ f^{n-1} - \frac{1}{\sum_{i < n} Df^i} \right) \circ h_n^{-1}. \quad (3.3.1)$$

Lemma 3.4.— The sum $\sum_{i < n} Df^i$ tends uniformly to ∞ .

Indeed, all the Df^i are positive, and infinitely many of them are $\geq e^{-V}$ (1.5). Applying this lemma to f and f^{-1} , we deduce from 3.4 that:

Lemma 3.5.— The Df_n converge uniformly to 1; it is therefore equivalent to prove that the V_n tend to 0, or that the variation of Df_n tends to 0.

Since variation is invariant under change of variable, we have

$$\begin{aligned} \text{Var}(Df_n) &= \text{Var}(Df_n - 1) \leq \text{Var}\left(\frac{Df}{\sum_{i<n} Df^{-i}}\right) + \text{Var}\left(\frac{1}{\sum_{i<n} Df^i}\right) \\ &= \left\| D \frac{Df}{\sum_{i<n} Df^{-i}} \right\|_1 + \left\| D \frac{1}{\sum_{i<n} Df^i} \right\|_1 \\ &\leq \left\| \frac{D^2 f}{\sum_{i<n} Df^{-i}} \right\|_1 + \|Df\|_\infty \left\| D \frac{1}{\sum_{i<n} Df^{-i}} \right\|_1 + \left\| D \frac{1}{\sum_{i<n} Df^i} \right\|_1. \end{aligned}$$

By 3.4, the function in the first term tends uniformly to 0. Theorem 3.1 therefore follows from the following result, applied to f and f^{-1} .

THEOREM 3.6.— If f is C^2 ,

$$D\left(\frac{1}{\sum_{i<n} Df^i}\right)$$

is uniformly bounded and tends almost everywhere to 0. Consequently, this function tends to 0 in the L^1 norm. Let G be the automorphism

$$\varphi \mapsto \varphi \circ f \cdot Df = f^*(\varphi dx)/dx$$

of $L^1(\mathbf{R}/\mathbf{Z})$. It preserves the L^1 norm and maps positive functions to positive functions. The Chacon-Ornstein ergodic theorem can therefore be applied to it (see, for example, Garsia, Topics in almost everywhere convergence): for every φ in L^1 , the quantities

$$\psi_n(\varphi) = \frac{\sum_{i=0}^{n-1} G^i(\varphi)}{\sum_{i=0}^{n-1} G^i(1)} = \frac{\sum_{i=0}^{n-1} \varphi \circ f^i \cdot Df^i}{\sum_{i=0}^{n-1} Df^i}$$

converge almost everywhere to a limit $\psi(\varphi)$.

We shall apply this theorem to $\varphi = D \log Df$. Put

$$\psi_n = \frac{\sum_{i<n} D \log Df \circ f^i \cdot Df^i}{\sum_{i<n} Df^i},$$

and denote by ψ the almost-sure limit of the ψ_n . The $|\psi_n|$, and hence $|\psi|$, are bounded above by $\|D \log Df\|_\infty$.

Lemma 3.7.—

$$D \left(\frac{1}{\sum_{i < n} Df^i} \right)$$

is uniformly bounded and tends almost everywhere (and hence also in the L^1 norm) to $-\frac{1}{2}\psi$.
With the summation indices running from 0 to $n-1$, we have

$$-D \left(\frac{1}{\sum Df^i} \right) = \frac{\sum D^2 f^i}{(\sum Df^i)^2} = \frac{\sum_{i > j} Df^i \cdot D \log Df \circ f^j \cdot Df^j}{(\sum Df^i)^2}.$$

The quantities

$$\lambda_{ij} = \frac{Df^i Df^j}{(\sum_{i < n} Df^i)^2} \quad (j < i < n)$$

are positive and have sum ≤ 1 ; hence one already sees that

$$-D \left(\frac{1}{\sum Df^i} \right) = \sum \lambda_{ij} D \log Df \circ f^j$$

is bounded in absolute value by $\|D \log Df\|_\infty$.

With the summation indices still running from 0 to $n-1$, we have

$$\begin{aligned} \sum_{i > j} Df^i \cdot D \log Df \circ f^j \cdot Df^j &= \sum_i Df^i \cdot \psi_i \cdot \sum_{j < i} Df^j \\ &= \psi \cdot \sum_{j < i} Df^i Df^j + \sum_i (\psi - \psi_i) Df^i \sum_{j < i} Df^j \\ &= \frac{1}{2} \psi \cdot \left(\sum Df^i \right)^2 - \frac{1}{2} \psi \sum (Df^i)^2 + \sum_i (\psi - \psi_i) Df^i \sum_{j < i} Df^j, \end{aligned}$$

and therefore

$$D \left(\frac{1}{\sum Df^i} \right) + \frac{1}{2} \psi = \frac{1}{2} \psi \cdot \frac{\sum (Df^i)^2}{(\sum Df^i)^2} + \frac{\sum_i (\psi - \psi_i) Df^i \sum_{j < i} Df^j}{(\sum Df^i)^2}.$$

Bound $\sum (Df^i)^2 = \sum Df^i \cdot Df^i$ by $\sup Df^i \cdot \sum Df^i$, and $\sum_{j < i} Df^j$ by $\sum Df^j$. This gives

$$\left| D \left(\frac{1}{\sum Df^i} \right) + \frac{1}{2} \psi \right| \leq \frac{1}{2} |\psi| \frac{\sup Df^i}{\sum Df^i} + \frac{\sum |\psi - \psi_i| Df^i}{\sum Df^i} = A_n + B_n.$$

For every j between 0 and $n-1$,

$$\frac{Df^j}{\sum Df^i} = \left(\sum Df^i \cdot (Df^j)^{-1} \right)^{-1} = \left(\sum Df^{i-j} \circ f^j \right)^{-1} \leq \inf \left(\frac{1}{\sum_{i < n-j} Df^i} \circ f^j, \frac{1}{\sum_{i \leq j} Df^{-i}} \circ f^j \right).$$

By 3.4, A_n therefore tends uniformly to 0. As for B_n , the following lemma, whose verification is left to the reader, shows that B_n tends to 0 at every point at which ψ_n tends to ψ .

Lemma 3.8.— Let a_n be a sequence tending to 0, and let (b_n) be the terms of a divergent positive series. Then

$$\frac{\sum_{i < n} a_i b_i}{\sum_{i < n} b_i}$$

tends to 0.

Finally, let us deduce 3.6 from 3.7. Since $1/\sum_{i < n} Df^i$ tends to 0,

$$D \left(1 / \sum_{i < n} Df^i \right)$$

tends to 0 in the sense of distributions. By 3.7, this sequence tends in L^1 to $-\frac{1}{2}\psi$. Hence $\psi = 0$, and 3.7 applies.

Supplement 3.9.— Herman can deduce from 3.7 that an increasing C^2 diffeomorphism of $\mathbf{R}/\mathbf{Z}$ with irrational rotation number is dx -ergodic: every Borel subset (indeed, every dx -measurable subset) of $\mathbf{R}/\mathbf{Z}$ that is invariant under f has Lebesgue measure 0 or 1.

Remark 3.10.— The Chacon-Ornstein theorem is not effective: it gives no estimate for the rate at which, for each ε , the ψ_n converge uniformly to ψ outside a set of measure ε . Consequently, I do not know how to deduce an effective bound for the V_n from the proof above.

§ 4. Approximation of a number by rationals

Except in 4.5, the general conventions concerning α, p, q are suspended in this section. We begin with some reminders about continued fractions, for which Hardy and Wright may be consulted.

4.1. Let $[a_0, \dots, a_n, \dots] = a_0 + 1/(a_1 + 1/(\dots (a_i \in \mathbf{Z}, a_i > 0 \text{ for } i \geq 1)))$

$i > 0$) be the continued-fraction expansion of an irrational number α . Let $p_n/q_n = [a_0, \dots, a_n]$ be the n th convergent. We have $p_n/q_n < \alpha$ for even n , and $p_n/q_n > \alpha$ for odd n . If, for $i = -1, -2$, (p_i, q_i) is defined by

$$\begin{pmatrix} p_{-2} & p_{-1} \\ q_{-2} & q_{-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

then for every $n \geq 0$,

$$\begin{pmatrix} p_{n-1} & p_n \\ q_{n-1} & q_n \end{pmatrix} = \begin{pmatrix} p_{n-2} & p_{n-1} \\ q_{n-2} & q_{n-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & a_n \end{pmatrix}, \quad (4.1.1)$$

that is,

$$\begin{cases} p_n = a_n p_{n-1} + p_{n-2}, \\ q_n = a_n q_{n-1} + q_{n-2}. \end{cases} \quad (4.1.2)$$

It follows from (4.1.2) that

$$a_{n+1} q_n \leq q_{n+1} \leq (a_{n+1} + 1) q_n, \quad (4.1.3)$$

$$2q_n \leq (a_{n+1} a_{n+2} + 1) q_n \leq q_{n+2}, \quad (4.1.4)$$

$$q_n \geq \sqrt{2}^n \quad \text{for } n \geq 2 \quad (\text{since } q_0 = 1, q_3 \geq 3 > 2^{3/2}). \quad (4.1.5)$$

By induction, (4.1.1) gives $p_{n+1} q_n - p_n q_{n+1} = \pm 1$, i.e.

$$\left| \frac{p_n}{q_n} - \frac{p_{n+1}}{q_{n+1}} \right| = \frac{1}{q_n q_{n+1}}. \quad (4.1.6)$$

Since α lies between p_n/q_n and p_{n+1}/q_{n+1} , it follows from (4.1.6) that the p_n/q_n are rational approximations to α , from below for even n and from above for odd n .

4.2. The following relations hold between convergents and rational approximations:

(4.2.1) Every rational approximation to s is either a convergent or is of the form $[a_0, \dots, a_{n-1}, a_n \pm 1]$. If the rational approximations are arranged in order of increasing denominators, then among any three successive ones at least one is a convergent.

(4.2.2) Of two successive convergents, at least one satisfies

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{2q^2}.$$

A rational approximation such that $|s - p/q| < 1/(2q^2)$ is a convergent.

4.3. Set

$$|||x||| = \inf_{n \in \mathbf{Z}} |x - n|.$$

The denominators q_n are the numbers q such that $|||q\alpha||| < |||q'\alpha|||$ for every $q' < q$. On $\mathbf{R}/\mathbf{Z}$, three successive points $q_n\alpha$ are arranged as follows (the drawing is for $a_{n+2} = 2$):

$$q_{n+1}\alpha \quad | \quad 0 \quad | \quad q_{n+2}\alpha \quad \underbrace{\hspace{1.5cm}}_{a_{n+2}(q_{n+1}\alpha)} \quad q_n\alpha. \quad (4.3.1)$$

If the case $n = 0$, $a_1 = q_1 = 1$, is excluded, then $|||q_n\alpha|||$ is the distance from 0 to $q_n\alpha$, as it appears in this figure, and

$$a_{n+2} \text{ is the integer part of } \frac{|||q_n\alpha|||}{|||q_{n+1}\alpha|||} \quad (\text{for } n \geq 1), \quad (4.3.2)$$

$$|||q_{n+2}\alpha||| \leq \frac{1}{a_{n+2} + 1} |||q_n\alpha||| \leq \frac{1}{2} |||q_n\alpha|||. \quad (4.3.3)$$

Set $p'_n/q'_n = [a_0, \dots, a_n + 1]$. The arrangement is

$$\frac{p_{n+1}}{q_{n+1}} \quad | \quad \alpha \quad | \quad \frac{p'_{n+1}}{q'_{n+1}} \quad | \quad \frac{p_n}{q_n}. \quad (4.3.4)$$

Hence, by (4.1.6) and (4.1.3),

$$\frac{1}{q_n q'_{n+1}} \leq \left| \alpha - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}}, \quad (4.3.5)$$

that is,

$$\frac{1}{(a_{n+1} + 2)q_n} \leq |||q_n\alpha||| \leq \frac{1}{a_{n+1}q_n} \quad (\text{except for } n = 0, q_0 = q_1 = 1). \quad (4.3.6)$$

Comparing (4.1.3) and (4.3.6), and taking (4.2.2) into account, one sees that the following requirements are equivalent:

- (a) $|||q\alpha|||$ is never too small;
- (b) a_n is never too large;
- (c) the denominators of the rational approximations to α do not grow too quickly.

DEFINITION 4.4.— An irrational number α has bounded density if its density

$$d = \sup_{n>0} \frac{1}{n} \sum_{i=1}^n a_i$$

is finite.

Apply 1.7 with $s = \alpha$ of density $d < \infty$, taking the q_i to be the denominators of the successive convergents. For $n \geq 2^{N/2}$, one has $N \leq q_n$ by (4.1.5), and

$$\sum_{q_i \leq N} a_{i+1} \leq nd.$$

Formula 1.7.3 gives the following result.

PROPOSITION 4.5.— There are universal constants C, D such that, when α has bounded density d ,

$$\|D^N f\|_\infty \leq C^{dV} \cdot N^{DdV}.$$

4.6. Let $T : [0, 1[\rightarrow [0, 1[$ send x to the fractional part of x^{-1} . The probability measure

$$\nu = \frac{1}{\log 2} \frac{dx}{1+x}$$

on $[0, 1[$ satisfies $T_*\nu = \nu$. Khintchine proved that T is ν -ergodic. The n -th coefficient in the continued-fraction expansion of an irrational $x \in [0, 1]$ is

$$a_n(x) = [(T^{n-1}(x))^{-1}] \quad (n \geq 1).$$

For almost every x (with respect to ν —or equivalently Lebesgue measure), an integer k therefore occurs in the sequence $a_n(x)$, $n \geq 1$, with frequency

$$\lim_{N \rightarrow \infty} \frac{1}{N} (\#\{n \leq N \mid a_n(x) = k\}) = \int_{[x^{-1}] = k} \nu \sim \frac{1}{k^2}.$$

Since $\sum_k (1/k^2) k$ diverges, the set of numbers of bounded density has Lebesgue measure zero.

Proposition 4.5 is the only step in Herman's proof at which one must impose on α a hypothesis that is not satisfied by almost every number.

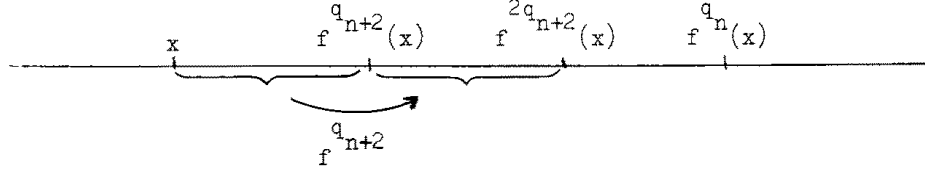
§ 5. Estimate of $|f^{q_n} - x - p_n|$

Write p_n/q_n for the n -th convergent of α . By 4.3.3, the following arrangement holds on the circle $\mathbf{R}/\mathbf{Z}$:



Authority-derived positional diagram P15-A01.

By conjugacy, the same arrangement holds for the iterates of f :



Authority-derived positional diagram P15-A02.

On $\mathbf{R}$, the same arrangement holds if f^{q_i} is replaced by $f^{q_i} - p_i$. The axis must be oriented from left to right for odd n , and from right to left for even n .

Since $Df^{q_i} \geq e^{-V}$, it follows that

$$(1 + e^{-V}) |f^{q_{n+2}}(x) - p_{n+2} - x| \leq |f^{q_n}(x) - p_n - x|,$$

and hence:

PROPOSITION 5.1.— The quantities $\|f^{q_n} - x - p_n\|_\infty$ decrease at least as fast as a geometric progression.

A slight refinement of this argument proves the following theorem.

THEOREM 5.2.— Suppose that α satisfies

$$(*) \quad \lim_{A \rightarrow \infty} \limsup_{N \rightarrow \infty} \frac{\sum_{\substack{a_i > A \\ 1 \leq i \leq N}} \log(a_i + 1)}{\sum_{1 \leq i \leq N} \log(a_i + 1)} = 0,$$

or, equivalently,

$$(*)' \quad \text{For every } \varepsilon \text{ there is } A \text{ such that } \prod_{\substack{a_i > A \\ 1 \leq i \leq N}} (a_i + 1) \leq O(q_N^\varepsilon).$$

There is then a function $V \mapsto \varepsilon(V)$, tending to 0 with V , such that

$$|f^{q_n}(x) - p_n - x| \leq O\left(q_n^{-1+\varepsilon(V)}\right).$$

Sketch of proof.

a) Let p/q and p'/q' be two rational approximations to α from above, and put $a = q\alpha - p$, $b = q'\alpha - p'$, and $c = a/b$. If $0 \leq rb \leq a$, where r is an integer ≥ 1 , then on $\mathbf{R}/\mathbf{Z}$ one has the following arrangement (drawn for $r = 3$):



Authority-derived positional diagram P16-A01.

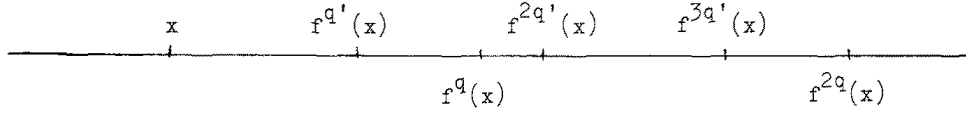
Since $Df^{q'} \geq e^{-V}$,

$$\left(\sum_{i=0}^{r-1} e^{-iV}\right) |f^{q'}(x) - p' - x| \leq |f^q(x) - p - x|.$$

b) More generally, if $rb \leq sa$, i.e. if $r/s \leq c$, then

$$\left(\sum_{i=0}^{r-1} e^{-iV}\right) |f^{q'}(x) - p' - x| \leq \left(\sum_{i=0}^{s-1} e^{-iV}\right) |f^q(x) - p - x|. \quad (5.2.1)$$

For $r = 3$, $s = 2$, the drawing is:



Authority-derived positional diagram P16-A02.

c) Given ε and A , there is B such that

$$\forall x (2 \leq x \leq A) \quad \exists r, s \quad (r \leq B, s \leq B, x^{1-\varepsilon} < r/s < x).$$

There is then V_0 such that, if $V < V_0$, for all $r, s < B$ with $r \geq 2s$,

$$\frac{\sum_{i=0}^{s-1} e^{iV}}{\sum_{i=0}^{r-1} e^{-iV}} \leq \left(\frac{s}{r}\right)^{1-\varepsilon}.$$

d) If $V < V_0$ and $2 \leq c \leq A$, (5.2.1) gives

$$\left|f^{q'}(x) - p' - x\right| \leq c^{-1+\varepsilon'} |f^q(x) - p - x|, \quad (5.2.2)$$

where $(1 - \varepsilon') = (1 - \varepsilon)^2$.

e) Take the sequence of rational approximations p_{2n+1}/q_{2n+1} , and set

$$b_n = |q_{2n+1}\alpha - p_{2n+1}| = |||||q_{2n+1}\alpha|||.$$

Then

$$2 \leq \frac{b_n}{b_{n+1}} \leq (a_{2n+2} + 1)(a_{2n+3} + 1).$$

If $V < V_0$, it follows that

$$\left\{ \begin{array}{ll} |f^{q_{2n+3}}(x) - p_{2n+3} - x| \leq \left(\frac{b_{n+1}}{b_n}\right)^{1-\varepsilon'} |f^{q_{2n+1}}(x) - p_{2n+1} - x| & (1) \\ \text{if } (a_{2n+2} + 1)(a_{2n+3} + 1) \leq A, & \\ |f^{q_{2n+3}}(x) - p_{2n+3} - x| \leq |f^{q_{2n+1}}(x) - p_{2n+1} - x| & (2) \\ \text{in every case.} & \end{array} \right.$$

Chaining these inequalities gives

$$|f^{q_{2n+1}}(x) - p_{2n+1} - x| \leq O\left(\left[\prod (a_{2i} + 1)(a_{2i+1} + 1)\right]^{1-\varepsilon'} \cdot b_n^{1-\varepsilon'}\right),$$

where the product is over $i \leq n$ such that $(a_{2i} + 1)(a_{2i+1} + 1) \geq A$.

It is bounded by the product $B_n = \prod (a_j + 1)^2$, taken over $j \leq 2n + 1$ such that $a_j + 1 \geq A^{1/2}$; also, b_n is bounded by $1/q_{2n+1}$. Thus

$$|f^{q_{2n+1}}(x) - p_{2n+1} - x| \leq O\left(B_n q_{2n+1}^{-1+\varepsilon'}\right).$$

The hypothesis on α allows B_n to be absorbed into ε' , which proves the result.

Remark 5.3.— Condition (*) holds for almost every number and for numbers of bounded density.

Remark 5.4.— Unless the a_i are bounded, this method unfortunately does not provide a lower bound for $|f^{q_n}(x) - p_n - x|$.

§ 6. Estimate of $|Df^{q_n} - 1|$

Let $\varphi_n = f^{q_n} - p_n - x$. We shall bound $|Df^{q_n} - 1| = |D\varphi_n|$ by Hadamard's interpolation formula (2.2.1). For α of bounded density and $f \in C^3$,

$$\|\varphi_n\|_\infty \leq O\left(q_n^{-1+\varepsilon(V)}\right) \quad \text{with } \varepsilon(V) \rightarrow 0 \text{ as } V \rightarrow 0 \quad (5.2),$$

$$\|D^2\varphi_n\|_\infty \leq O\left(q_n^{1/2} \cdot \sup_{i < q_n} \|Df^i\|_\infty\right) \quad \text{for } V < 1 \quad (2.3),$$

$$\sup_{i < q_n} \|Df^i\|_\infty \leq O\left(q_n^{\varepsilon'(V)}\right) \quad \text{with } \varepsilon'(V) \rightarrow 0 \text{ as } V \rightarrow 0 \quad (4.5).$$

Consequently,

$$\|D\varphi_n\|_\infty \leq O\left(q_n^{-1/4+\varepsilon''(V)}\right), \quad \varepsilon''(V) = \frac{1}{2}(\varepsilon(V) + \varepsilon'(V)).$$

For every $\varepsilon > 0$, there is m such that

$$\varepsilon''(\text{Var log } Df_m) < \varepsilon \quad (3.3),$$

and hence

$$\|Df_m^{q_n} - 1\|_\infty \leq O\left(q_n^{-1/4+\varepsilon}\right).$$

By conjugacy ($f^{q_n} = h_m f_m^{q_n} h_m^{-1}$, $f_m^{q_n}$ is close to $\mathbf{R}/\mathbf{Z}$ by 5.2, and h_m is C^3), we finally obtain:

THEOREM 6.1.— If f is C^3 and α has bounded density, then for every $\varepsilon > 0$,

$$\|Df^{q_n} - 1\|_\infty \leq O\left(q_n^{-1/4+\varepsilon}\right).$$

§ 7. h is $C^{1+\beta}$ for $\beta < 1/3$

7.1. We shall use the following properties of the Hölder seminorm $|\varphi|_\beta$.

(7.1.1) If φ has period 1 and $g : \mathbf{R}/\mathbf{Z} \rightarrow \mathbf{R}/\mathbf{Z}$, then

$$|\varphi \circ g|_\beta \leq |\varphi|_\beta \|Dg\|_\infty^\beta.$$

(7.1.2) $|\varphi|_\beta$ is an increasing log-convex function of β .

Indeed, $\log |\varphi|_\beta$ is the supremum of the increasing linear functions

$$\log \frac{|\varphi(x) - \varphi(y)|}{|x - y|^\beta}.$$

$$|\varphi|_0 \leq 2\|\varphi\|_\infty \quad \text{and} \quad |\varphi|_1 = \|D\varphi\|_\infty. \quad (7.1.3)$$

$$\|D\varphi\|_\infty \leq \left(\frac{1+\beta}{2\beta}\right)^{\beta/(\beta+1)} \cdot \left(|\varphi|_0^\beta |D\varphi|_\beta\right)^{1/(1+\beta)}. \quad (7.1.4)$$

Suppose that the maximum of $|D\varphi|$ is attained at x_0 , and, to fix ideas, that $D\varphi(x_0) \geq 0$. Then

$$D\varphi(x) \geq \|D\varphi\|_\infty - |D\varphi|_\beta(x - x_0)^\beta$$

and

$$\varphi(x) - \varphi(x_0) \geq \|D\varphi\|_\infty(x - x_0) - |D\varphi|_\beta \frac{(x - x_0)^{1+\beta}}{1 + \beta}, \quad x \geq x_0.$$

Evaluate this lower bound at x such that $\|D\varphi\|_\infty - |D\varphi|_\beta(x - x_0)^\beta = 0$. Combining it with the symmetric result for $x < x_0$, one obtains

$$\frac{1}{2}|\varphi|_0 \geq \|D\varphi\|_\infty X - |D\varphi|_\beta \frac{X^{1+\beta}}{1 + \beta} = X \left(\|D\varphi\|_\infty - \frac{|D\varphi|_\beta X^\beta}{1 + \beta} \right)$$

for X such that $\|D\varphi\|_\infty = |D\varphi|_\beta X^\beta$. Formula (7.1.4), of which Hadamard's formula (2.2.1) is the special case $\beta = 1$, follows.

(7.1.5) $\|\varphi\|_\beta = \|\varphi\|_\infty + |\varphi|_\beta$ is a Banach-algebra norm on the algebra of C^β functions. The formula $\|\varphi_1\varphi_2\|_\beta \leq \|\varphi_1\|_\beta \|\varphi_2\|_\beta$ follows from the printed identity

$$\varphi_1(x)\varphi_2(x) - \varphi_1(y)\varphi_2(y) = \varphi_1(x)(\varphi_2(x) - \varphi_2(y)) + (\varphi_1(x) - \varphi_1(y))\varphi_2(y).$$

7.2. Assume that α has bounded density and that $f \in C^3$. Let $\varepsilon < 1/4$. By 6.1,

$$|\log Df^{q_n}| \leq A \cdot q_n^{-\varepsilon}. \quad (7.2.1)$$

Let N be an integer and write $N = \sum b_i q_i$ as in 1.7. Breaking the sum

$$\log Df^N = \sum_{i < N} \log Df \circ f^i$$

into partial sums as in 1.7 gives

$$\log Df^N = \sum_i \left(\sum \text{of } b_i \text{ terms of the form } \log Df^{q_i} \circ f^j \right). \quad (7.2.2)$$

Applying (7.2.1):

$$\|\log Df^N\|_\infty \leq A \sum_{i=0}^{\infty} a_{i+1} q_i^{-\varepsilon}. \quad (7.2.3)$$

This series converges if α has bounded density (and for almost every number), hence:

PROPOSITION 7.3.— $\|\log Df^n\|_\infty$ is uniformly bounded.

7.4. Suppose $V < 1$, and interpolate by 7.1.2 between the estimate in 6.1 for $\|Df^{q_n} - 1\|_\infty$ and the estimate in 2.3 for $\|D(Df^{q_n} - 1)\|_\infty$. For $\beta < 1/3$, this gives

$$|Df^{q_n}|_\beta \leq O(q_n^{-\varepsilon}) \quad \text{with } \varepsilon > 0. \quad (7.4.1)$$

Take $|\cdot|_\beta$ in (7.2.2). The compositions by f^j are harmless by (7.1.1) and 7.3, so

$$|Df^n|_\beta \leq O\left(\sum a_{i+1} q_i^{-\varepsilon}\right) = O(1) :$$

the f^n are uniformly $C^{1+\beta}$, as are the h_n , and therefore h is $C^{1+\beta}$, as explained in the introduction. Finally, 3.3 removes the hypothesis $V < 1$.

PROPOSITION 7.5.— For every $\beta < 1/3$, h is $C^{1+\beta}$.

§ 8. $C^{2+\beta}$ convergence of the f_n

THEOREM 8.1.— Let $0 < \beta' < \beta \leq 1$. Suppose that f is $C^{2+\beta}$ and h is $C^{1+\beta}$. Then the f_n converge to R_α in the $C^{2+\beta'}$ topology.

The convergence is already known to be C^1 (3.5). Here this is also immediate from (3.3.1), since the Df^i are bounded below:

$$Df^i > A > 0.$$

The interpolation formulas (7.1.2) and (7.1.4) therefore reduce the problem to showing that the f_n remain bounded in the $C^{2+\beta}$ topology, i.e. that the $|D^2 f_n|_\beta$ remain bounded.

The f^n are uniformly $C^{1+\beta}$. Thus

$$\left| \frac{1}{n} \sum_{i < n} Df^i \right|_\beta$$

is bounded. Since the Df^i are bounded below, and

$$|\varphi^{-1}|_\beta \leq \frac{|\varphi|_\beta}{(\inf |\varphi|)^2},$$

we also have boundedness of

$$\left| \frac{n}{\sum_{i < n} Df^i} \right|_{\beta}.$$

By the same method as in 3.6, using the fact that the h_n and f^n are uniformly C^1 (which justifies the changes of variables), it remains to bound uniformly in the norm $\|\cdot\|_{\beta}$ of (7.1.5) the functions

$$-D \left(\frac{1}{\sum_{i < n} Df^i} \right) = \sum_{j < i < n} \lambda_{ij} D \log Df \circ f^j$$

where

$$\lambda_{ij} = \frac{Df^i Df^j}{(\sum_{i < n} Df^i)^2} = \frac{1}{n^2} \left[Df^i \cdot Df^j \cdot \left(\frac{n}{\sum_{i < n} Df^i} \right)^2 \right] \quad (\text{proof of (3.7)}).$$

The expression in brackets and the functions $D \log Df \circ f^j$ are uniformly bounded in the norm $\|\cdot\|_{\beta}$, so this is immediate.

§ 9. Application of an implicit-function theorem

Herman's final result is the following.

THEOREM 9.1.— If α has bounded density and f is C^r , $r \geq 3$ ($r = n + \beta$, with integer n and $0 \leq \beta < 1$), then h is $C^{r-1-\varepsilon}$ for every $\varepsilon > 0$.

He obtains it by combining 7.5 and 8.1 with an implicit-function theorem:

THEOREM 9.2.— Let α be irrational and let $1 \leq \theta < \theta'$. Suppose there is C such that

$$|q\alpha - p| \geq C \cdot q^{-\theta}$$

for all integers p, q . There is then a neighborhood V of R_{α} in the $C^{2\theta'}$ topology such that, for f in this neighborhood and of class C^r (respectively C^{∞} , C^{ω}), the equation

$$f = R_{\lambda} \circ g \circ R_{\alpha} \circ g^{-1}$$

with $\lambda \in \mathbf{R}$ and g an increasing homeomorphism has a $C^{r-\theta'}$ solution (respectively a C^{∞} , C^{ω} solution).

To obtain 9.1, take $1 < \theta < \theta' = 1 + \varepsilon$, take α to be the rotation number of f , reduce by 7.5 and 8.1 to the case where $f \in V$, and observe that the rotation number of

$$R_\lambda \circ g \circ R_\alpha \circ g^{-1}$$

can equal α only when $\lambda = 0$.

As already stated, the proof of 9.2 is inspired by [1], [3], and [4].

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